COMP4702 Report

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1 Introduction

1.1 Objectives

The first objective of this report is to compare the results between using the K-nearest neighbours (KNN) algorithm and logistic regression, when categorising a player's gender based on the acceleration of the player's swing. The second objective of this report is to implement linear regression to investigate any potential linear positive relationship between the maximum acceleration of a player's swing, and the mean and variance of the x, y, and z axes accelerations.

2 Data Cleaning

2.1 Data Reduction

The given dataset has 51 features. Focusing on our objectives, some of these features are values that we do not need. We remove all columns containing data about the angular velocity of a player's swing as we are not exploring angular velocity at all in this report. We also remove the following columns that are unnecessary for our report: date, age, weight, playYears, height, newv1, newv2, newv3, and newv4.

2.2 Null values, duplicates, and outliers

We remove all rows whose acceleration values or weight are equal to null. This is because null values can imply an invalid measurement of data or a data entry mistake. We also remove rows whose acceleration values are not a number (NaN). We further remove rows of duplicates. We do not keep any occurrence as we do not have the knowledge as to whether these duplicates are coincidences or mistakes. This prevents our model from learning incorrect patterns, preventing inaccuracy and incorrect conclusions. Outliers are not removed as no reason has been given to believe that those values have been measured incorrectly.

3 K-nearest Neighbours

3.1 Implementation

We split the data such that 80% of it is for training, and 20% is for testing. Here, we focus on the following features: ax_mean, ay_mean, az_mean, ax_var, ay_var, az_var, ax_rms, ay_rms, az_rms, a_max, a_mean, a_min, a_fft, a_psdx, a_kurt, a_skewn, a_entropy, and gender. Leaving gender as is, we scale all of the data such that all values are between -1 and +1. This is because KNN is an algorithm that computes distance and is thus known to be rather sensitive to the magnitude of feature values.

3.2 Training

We experimented with different values for the number of neighbours, ranging from 3^0 to 3^{10} . For the distance metric, we chose to use Euclidean distance as our data is continuous and numerical. We found that, as k-neighbours increase from 1 to 5, the accuracy of our KNN model improves. Our best accuracy score is at k=5, with accuracy= 0.87777 (5 decimal places). However, from 6 all through till 3^{10}, the accuracy of our KNN model deteriorates rather quickly.

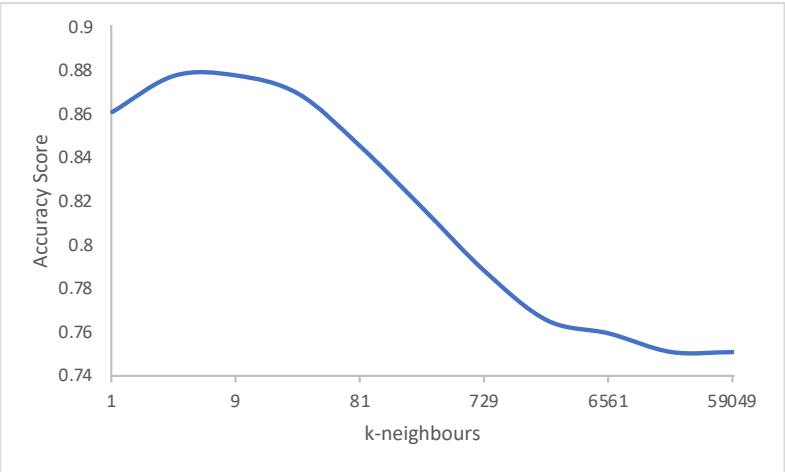

Figure 1: KNN model's accuracy

3.3 KNN Discussion

I consider our findings to be quite contradictory. This is because theoretically speaking, KNN tends to perform better with a moderately-sized k (relative to the number of data points). In our context, however, the model's performance was far superior with small values of k. Upon reflection, this could be due to the fact that there exist overlaps between the classes in our data. If many men and women recorded in our datasheet produce swings of similar acceleration, smaller k values will be more sensitive in identifying minuscule variations. This results in better accuracy.

To further evaluate our model, we implement the confusion matrix. We can see that the overall accuracy is rather balanced in the sense whereby the model is not misclassifying one class drastically more than the other. That being said, however, our KNN model did subtly worse when classifying females. As mentioned earlier, this could be due to the fact that there exist numerous females and males in our data that have very similar (or the same) accelerations of swing. Our model could be assuming these females to be male. This, in turn, could be due to the male class dominating the dataset. It can also be because of outliers in our data whereby females have larger accelerations than their counterpart, though it is usually the opposite.

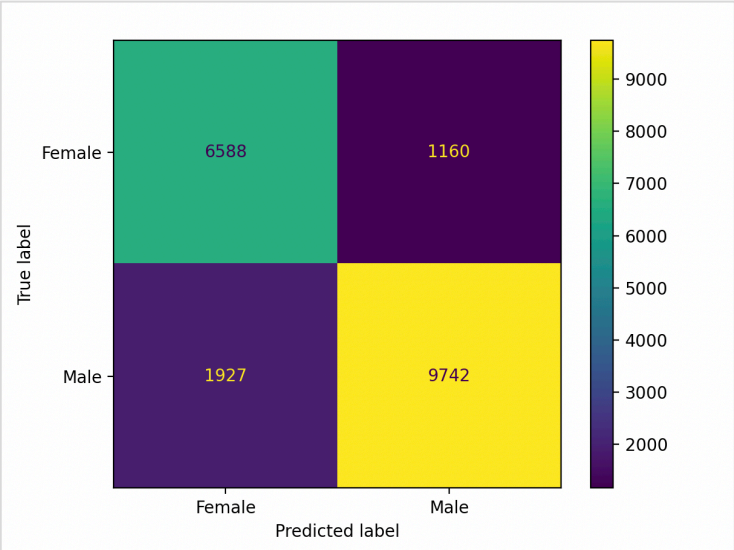

Figure 2: Confusion Matrix for KNN classifier, k=4

4 Logistic Regression

4.1 Implementation

Similar to our KNN model, we split the data such that 80% of it is for training, and 20% is for testing. We focus on the exact same features as listed in section 3.1. The data remains scaled the same way we did for our KNN model. This ensures that all features have equal contributions to our classifier's predictions regardless of their individual scales. This is also necessary for us in order to perform regularisation on our logistic regression model.

4.2 Training with L2 Regularisation

With L2 regularisation, our logistic regression model was able to obtain an accuracy score of 0.63892 (5 decimal places). This was obtained with the inverse of regularisation strength (C) being 1. We then experiment with different values for C. As we ever so slightly increase C, our model's accuracy quickly declines. Thus, we focus on lowering the value of C. We found that, as C decreases, the accuracy score of our model also worsens (but to a much less extent).

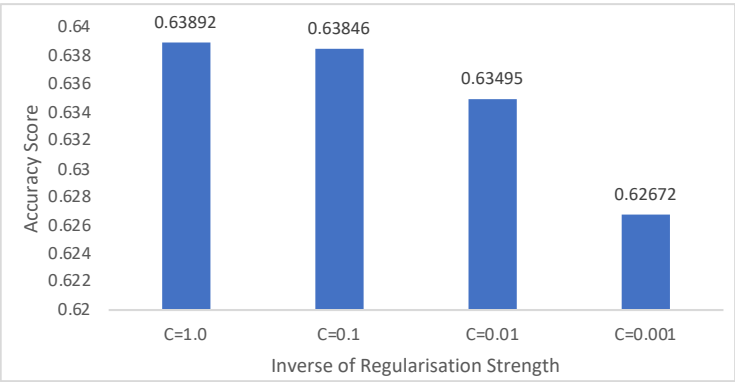

Figure 3: Accuracy score of logistic regression model with respective C values

Upon reflection, this could be due to potential overlap between classes in our dataset. This aligns with our discussion for KNN in section 3.3. Smaller C values smoothen the decision boundaries too much, resulting in more misclassifications. Conversely, larger C values led to the overfitting of our model. Hence, we found that we have the best accuracy at C=1.0.

4.3 Logistic Regression Discussion

I think that our findings are able relate to a theoretical aspect of logistic regression. This is pertaining to how logistic regression assumes a linear decision boundary. As inferred in section 3.3, overlaps between the female class and the male class in our dataset are likely to exist. The decision boundaries of our data are thus extremely likely to be blurred and complex. I think that our logistic regression model is too simple and thus incapable of representing complexities in our dataset. This would likely be the main reason why we only managed to obtain an accuracy score of 0.63892 (5 decimal places).

To further evaluate our model, we implement the confusion matrix. As we can see from the figure below, the model is misclassifying females significantly more than the other. Apart from our model being too simple to capture our dataset's complexities, there could also be a possibility that the female class is underrepresented as compared to the male class. As logistic regression is known to be sensitive to class imbalances, our model might have favoured the male class more if our dataset is dominated by the male class.

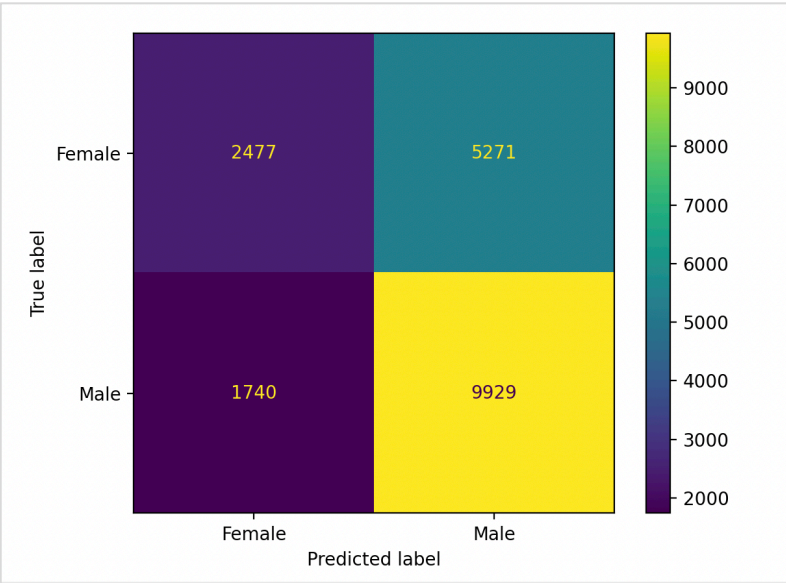

Figure 4: Confusion Matrix for Logistic Regression Model, C=1.0

5 K-nearest Neighbours & Logistic Regression Comparison

5.1 Cross Validation & Discussion

We use k-fold cross validation to assess the performance of our KNN and logistic regression models. We experiment with multiple different number of splits, ranging from 5 to 10. Looking at the cross validation scores, we found that the resulting average accuracy of our KNN model is worse than the accuracy score we managed to obtain with train test split. The highest accuracy score for our KNN model through cross validation is only 0.50204 (5 decimal places) at 8 splits. However, for our logistic regression model, the results from cross validation were similar (though worse, too) to that of its train test split results. The highest accuracy score for our logistic regression model through cross validation is 0.62274 (5 decimal places) at 7 splits.

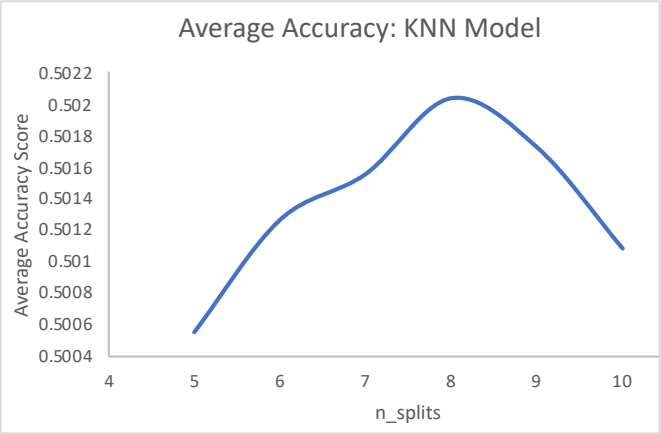

Figure 5: KNN average accuracy score using K-fold cross validation

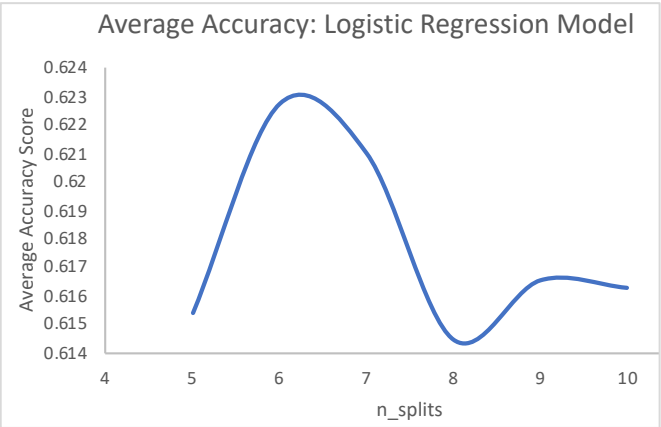

Figure 6: Logistic regressions average accuracy score using K-fold cross validation

I think that the drastic drop in our KNN accuracy score could be due to class imbalance. This would make sense as our K-fold evaluation would favour the majority class. This aligns with

discussions made in both sections 3.3 and 4.3, where we pointed out the possibility of an imbalance in the classes of our dataset. To finally test this theory, I ran a code to determine the proportion of the female and male class in our dataset. As speculated, 39.7% of the samples in our dataset are female and 60.3% are male.

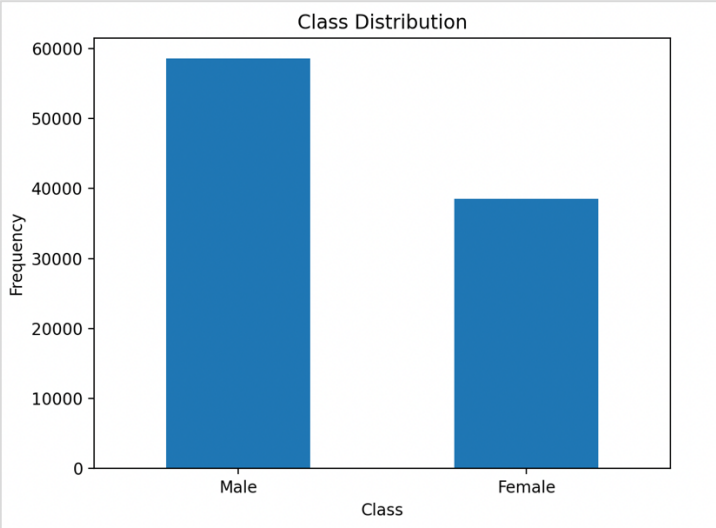

Figure 7: Proportion of each class in our dataset

5.2 Comparison Conclusion

I personally think that our logistic regression model outperformed our KNN model. This is because, although KNN obtained much higher accuracy scores when train test split was applied, its average accuracy score after cross validation is similar to that of randomly guessing. This just implies poor generalisability. Conversely, though our logistic regression model only achieved an accuracy score of ~63 for train test split, its score remains consistent with cross validation. This suggests stability and potential for better generalisability.

In addition, we get to confidently conclude that there is certainly a class imbalance in our dataset where the male class dominates the female class. We have also further inferred that there exists overlaps between the male and female classes in our dataset, as well as complex relationships in our data. We can thus conclude that there does not exist a linear class separation in our data between the acceleration of a male's swing and that of a female's swing.

6 Linear Regression

6.1 Implementation

We chose not to standardise our data. We split the data such that 80% of it is for training, and 20% is for testing. We only focus on the following features: ax_mean, ay_mean, az_mean, ax_var, ay_var, az_var, and a_max. To ensure that our input variables are not multicollinear, we implement a correlation matrix and subsequently a heatmap.

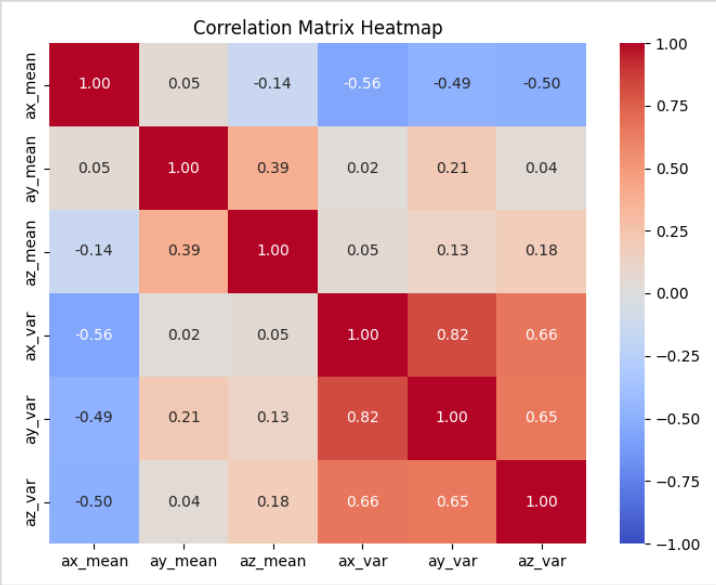

Figure 8: Correlation matrix heatmap of our input variables

We found that there is a very high correlation (of more than 0.8) between ax_var and ay_var. We attempt the correlation matrix and its respective heatmap again ,without the ax_var feature.

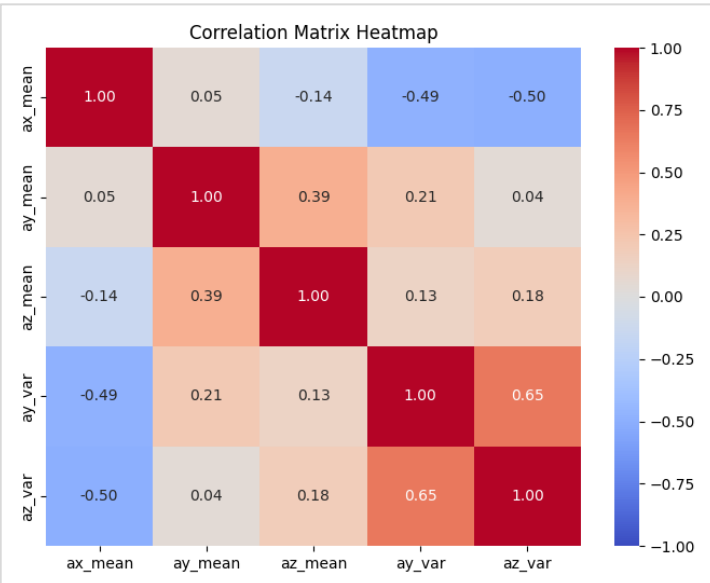

Figure 9: Correlation matrix heatmap of our input variables without ax_var

6.2 Training

Without regularisation, we obtained an R^2 value of 0.86786 (5 decimal places). Regardless of L1 regularisation or L2 regularisation, our R^2 value also remained the same. Even as we vary alpha, our model performs the same. Thus, we choose to remove regularisation from our code. We derived the following regression diagnostics plot and found that our points lie very close to y=x. This implies that our model's predicted values are rather close to the actual values.

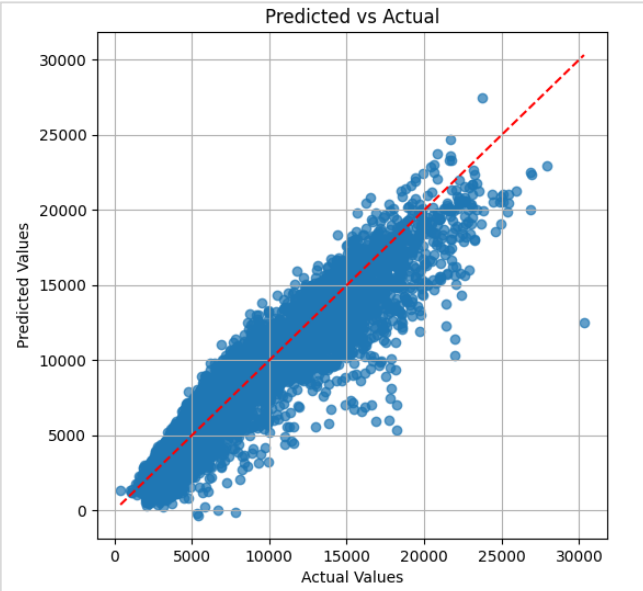

Figure 10: Regression diagnostics plot without regularisation

We then plot our residuals. We found that our residuals are generally randomly scattered around the red line with (however) more points above the red line than there are below.

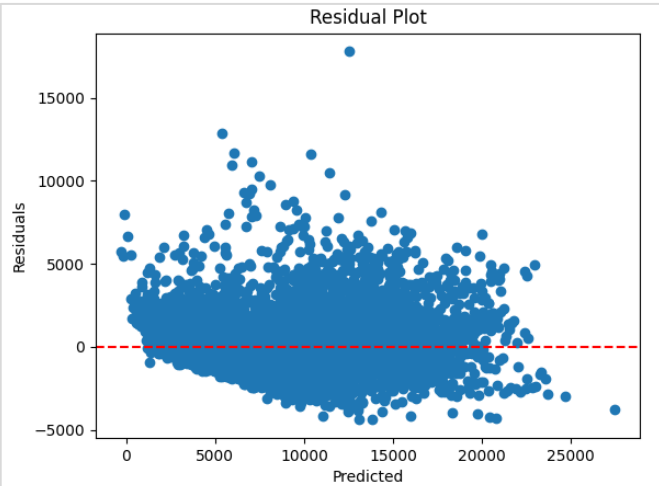

Figure 11: Residual plot without regularisation

6.3 Linear Regression Discussion

Given that regularisation did not improve our model, we contemplate multiple possibilities to explain this. Firstly, it cannot be an issue with the regularisation strength as we have experimented with a wide range of alpha values but our model's results remain relatively similar. We move on to the next possibility, which is that our data is already clean and have low correlation with one another. This is likely to be a contributing factor since it aligns with our correlation matrix in section 6.1. Lastly, it could be that the number of features we use for the model is small. This is likely to be the other contributing factor as we only use five features as the input variables to our model.

We now discuss the results of our residual plot (Figure 11) in section 4.2. We reflect upon the fact there are more points above the red line than below it. This can imply that our model is overestimating. However, this cannot be the case since regularisation did not improve our model. This can also imply a non-linear relationship between our variables, but this cannot be true either as our regression diagnostic plot (Figure 10) in section 4.2 follows y=x, implying that a linear relationship exists. The most plausible explanation is that we have outliers that are actively skewing our model. This theory aligns with the way our dataset was derived, that we are handling experimental values of a human's swing. There are, hence, bound to be outliers that I (personally) think should not be "treated" nor removed as it would mean intentionally manipulating our data to achieve an artificially improved model performance.

7 Conclusion

Firstly, we deduced that there exist overlaps between the female class and the male class in our given dataset. We also inferred that relationships between features present in our dataset are complex. This is because, with smaller k values, our KNN classifier was able to pick up on smaller variations. Subsequently, through the comparison of our KNN classifier and our

logistic regression model in section 5, we are also able to deduce that there is an imbalance between the female class and the male class in our dataset. The male class dominates the female class with a ~60:30 ratio. We then concluded that there does not exist a linear class separation between the acceleration of a male's swing and the acceleration of a female's swing. Next, we observed a positive and linear relationship between the maximum acceleration of a player's swing, and the mean of the x, y, and z axes accelerations, as well as the variance of the y and z axes. We, however, conjecture that there are outliers present in our data that skew our linear regression model. Despite positive results, we are then able to infer that a human's swing cannot be perfectly modelled through linear regression.